

Analysis of Soil cesium levels and Pediatric thyroid cancer incidence rates in the south western region of Belarus in 1992 – Chernobyl Disaster.

Jyothi sri Lokanadham,

Final Paper,

Environmental Data science course,

Colorado School of Public health.



Getty/CIA factbook 1996 Chernobyl map

Spatial Analysis of the soil Cesium(^{137}Cs) levels and Pediatric thyroid cancer – Chernobyl Disaster:

Abstract: Chernobyl disaster happened in the year 1986, in the city of Pripjat, Ukraine, which resulted in the release of radioactive materials in huge amounts for nearly ten days, the effects of which can be witnessed till date. Among the radioactive release, material with shorter half-life such as radioactive Iodine and Cesium are accounted for most of the deleterious aftereffects such as genetic defects and cancers. The current study aims to analyze the soil ^{137}Cs levels in the region of Belarus and its correlation with the pediatric thyroid cancers in the year 1992. The results obtained show a strong correlation between the levels of soil ^{137}Cs levels and the incidence rate of pediatric thyroid cancers, with an increase of nearly 1.99 cases per 1000 children on an average, for every one curie per kilometer square increase in ^{137}Cs , in the year 1992 in the southwest region of Belarus. (P value- 7.56×10^{-78})

Introduction:

The Chernobyl disaster occurred in the year 1986, 26th of April 1:23am at the RBMK 1000 - design reactor no.4, just 2.5 years of its first operation in 1983.(1) Though Xenon poisoning and reduction of the power to lower than recommended limits are cited as significant contributors during the accident, it should also be noted that change in the shift of the work force at 12:00am during a running reactor test, lack of notable technical and operational improvements in spite of failure of the previous three tests, faulty RBMK design with respect to the length of the graphite rods, serious operational faults – all these have contributed to the result of an explosion at the Reactor 4, Chernobyl site resulting in massive radioactive plumes that spread across Europe in the coming days.(1) The explosion was caused due to building up of steam and xenon gases due to extreme heat, thus blowing up of the 1000 ton steel top and releasing the radioactive materials into the air. Nearly 60% of the radioactive deposits occurred in Belarus due to the north-west direction of the wind. Immediate challenges included localization of the fire and prevention of its spread to reactor 3, by using nearly 200 to 300 tons of water jets per hour, to cool the graphite. This was followed by dropping of neutron absorbing materials such as lead and boron at the disaster site to control the spread of the radioactive materials.(1) Nearly 600,000 liquidators were deported to the site to manage the situation, from years 1986 to 1989 with each one of them receiving nearly 100 milli sieverts of radiation. (2)The disaster resulted in 30 immediate onsite deaths followed by nearly 237 people onsite involved in the cleanup operations reported with Acute radiation syndrome. Nearly 14×10^8 becquerel of radioactivity was released into the air resulting in severe long-term effects for years to come, of which the short-lived radioactive iodine and cesium proved to be fatal resulting in nearly 20,000 thyroid cancers in the years 1991 to 2015, among patients exposed to radiation below 18 years of age.(3) In this study, the incidence rates of pediatric thyroid cancer in the year 1992 in the region of Belarus and soil ^{137}Cs levels are analyzed to determine any existing association.

Aims and Hypothesis:

The current spatial analysis aims to correlate the soil ^{137}Cs levels in the southeastern region of Belarus to the incidence rate of Pediatric thyroid cancers in the year 1992. In addition, the study aims to determine the correlation between soil ^{137}Cs levels and berry cesium levels for the year 1992. It is hypothesized that the incidence rates of pediatric thyroid cancer are higher in areas closer to the Chernobyl site.

Methods:

It is a secondary spatial data analysis performed using R 4.2.2 software. Soil cesium levels for a 60-kilometer radius from the Chernobyl reactor 4 site in the year 1987, has been obtained from the Environmental information data centre, UK (Valery Kashparov, Sviatoslav Levchuk, Marina Zhurba, Valentyn Protsak, Nicholas A. Beresford, and Jacqueline S. Chaplow, Earth System Science Data. 2019). The soil sampling sites included a total of 540 selected points of which after exclusion of rivers and swamps included a total of 489 sampling points. Soils were sampled using a corer with a diameter of 14 cm and depth of 5 cm. Equivalent dose rate was measured at a height of 1 m above ground at each sampling point. At each sampling site, soil samples were collected from five points using an envelope sampling design (with approximately 5-10 m between sampling points). Each soil sample was retained intact within the corer during transportation to maintain sample integrity and prevent mixing. (All the information stated here has been obtained from the supporting documentation provided with the dataset) The soil samples were analysed for gamma emitting radionuclides – ⁹⁵Zirconium, ⁹⁵Neodymium, ¹⁰⁶Ruthenium, ¹³⁴ and ¹³⁷Caesium and ¹⁴⁴Cerium. (The links to the data are provided below – Appendix 1) The distance variable in the dataset is changed from kilometres to metres.

Data regarding the incidence rates of Pediatric thyroid cancer (1985 to 1995) has been obtained from the International Atomic Energy Agency. (The links to the data are provided below – Appendix 1)

Soil Cs137 levels for the country of Belarus, Cs137 levels in the berries (1 kilogram of berries from the forest sample sites) and thyroid cancer incidence rates, target population estimates in the districts of Belarus are obtained from ESRI datasets (1992). (The links to the data are provided below – Appendix 1)

Pulkovo 1942 zone 5 projection has been used for all the map projections.

Soil Cs137 levels at 60 km radius around the Chernobyl site are mapped to evaluate the soil Cs137 levels in the year 1987.

Soil Cs137 levels for the country of Belarus for the year 1992 are used to predict the levels from years 1986 to 1996. This data unfortunately was not used for estimation of thyroid cases due to limitations in the coordinate system and projections which were not revealed in the dataset or supporting documentation. Hence, dataset including the south western region of Belarus is used to build a linear model to understand the association between soil Cs137 levels and thyroid cancer incidence rates for the year 1992.

Statistical Analyses:

All the datasets are read into the R environment and are projected into Pulkovo 1942 zone 5 crs. Primary variables of interest include – Soil Cs137 levels in 1987 and 1992 in the region of Belarus, Incidence rates of Thyroid cancer, distance from the Chernobyl site, and Berry Cs137 levels. Soil, berry Cs137 levels are converted into Curie per km². Dataset comprising of the soil Cs137 levels within 60 km radius around the Chernobyl site contained angle and distance from the Chernobyl site. The Chernobyl location coordinates which are available in another dataset are used along with the angle values and the distance variable, to obtain the x and y coordinates of the spatial points of Cs137 in soil. In order to understand the Cs137 levels near the Chernobyl site in 1987, the dataset with soil Cs137 levels within 60 km radius has been mapped, along with the Chernobyl site location point. The dataset with the incidence rates of pediatric thyroid cancer is used to calculate the relative

risk, and expected thyroid cases using probmap function from the sdep package. Incidence rates of thyroid cancer for the year 1992 has been mapped. The dataset with soil Cs137 levels is joined with the thyroid cancer polygons dataset using st_intersection restricting only to areas where the soil Cs137 values are available and mapped. The thyroid polygons are then joined to the berry Cs137 dataset and the spatial points are mapped. Finally, both the datasets are joined using st_geometry, thus retaining the thyroid dataset polygons which contain the district boundaries. Polygons with missing soil Cs137 and Berry Cs137 are removed. A linear regression model was built to understand the association between soil Cs137 levels and berry Cs137 levels. A linear regression model was built to look at the correlation between the soil Cs137 levels at different sites and the distance of the sites from the Chernobyl site.

Results:

The map representing the soil Cs137 values within 60 km radius in 1987 shows high Cs137 levels greater than 200 curie per kilometer square near the border of Belarus and above than when compared to the values southwards, thus emphasizing the influence of the winds in northwestern direction. (map1) A sharp increase in the average incidence rate(per 100,000) of pediatric thyroid cancers can be seen from 1986 and the number of cases follow an increasing trend until 1992. (Table1) The incidence rates seem to be higher in girls when compared to boys. The map represents a higher incidence rate(per 1000) in the districts of Belarus closer to the Chernobyl site with nearly 2 cases per 1000. (Map2) A higher levels of Cs137 in berries has also been observed in areas closer to the Chernobyl site. On running a linear regression model, for every one curie/kilometer square increase in soil levels Cs137 levels, there has been a statistically significant increase in the incidence of pediatric thyroid cancer by nearly 1.99 cases per 1000 children on an average in the year 1992 in the south west region of Belarus. (P value= 7.56×10^{-78}) (Table2) For every one curie/kilometer square increase in soil levels of log cs137, the concentration of Cs137 in the berries significantly increases by 1.49×10^{-10} curie per kilogram of berries. (P value= 5.17×10^{-15}) (Table3) For every one kilometer increase in distance from the Chernobyl site, the soil Cs137 values decreased by 1.34×10^{-6} curie/km².

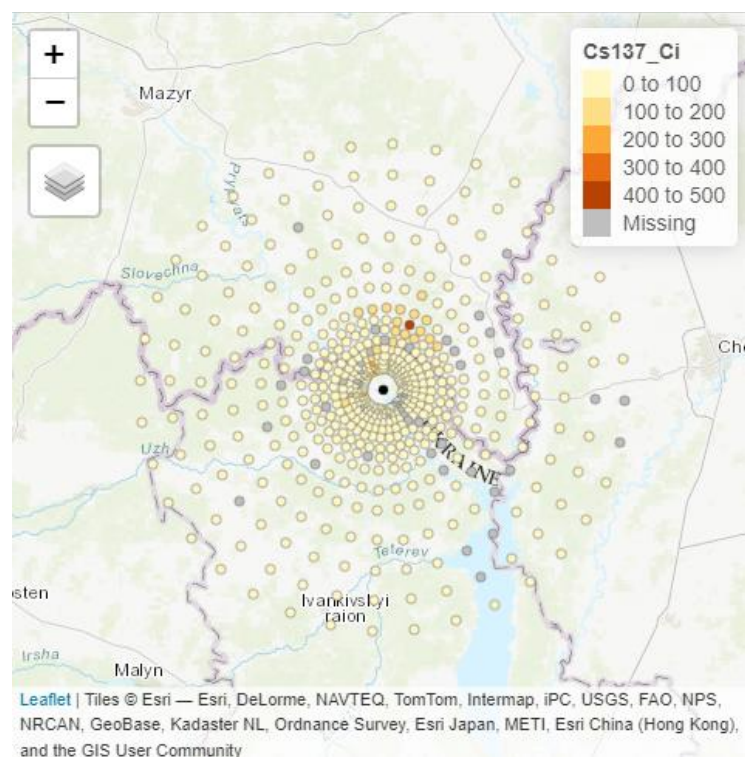
Discussion:

According to the US environmental protection agency, nearly 200 Pico curie of Cs137 per liter of water is generally considered as the annual permissible limit(4). Food grown in soils with high levels of Cs137 and water are the main routes of Cs137 entry into the body.(4) (5) The soil samples of Cs 137 ranged from 0 to 500 curie per km² within the 60km radius in 1987. Over the years though the amount of soil Cs137 reduced, values ranging from 0 to 140 curie per km² can be noticed in 1992, which is far from the safer levels. The actual number of deaths that can be attributed to the Chernobyl disaster is controversial till date.(6) Though immediate management efforts have been undertaken effectively by USSR to control the fires at the Chernobyl site and reduce the release of radioactive materials, there have been serious flaws in reducing the exposure to the populations living near the Chernobyl Site and in Belarus. One of the significant factors that resulted in the absorption of Cs137 and pediatric thyroid cancers is the severe unattended shortage of Iodine pills for children which would have helped in significant reduction in the Cs137 by thyroid gland and also high levels of Cs137 in milk which was consumed mostly by children.(7) The Chernobyl disaster serves as a stark illustration of the devastating consequences that can arise from a combination of severe technical flaws in the RBMK reactor, human error leading to critical mistakes, and inadequate management of the aftermath.

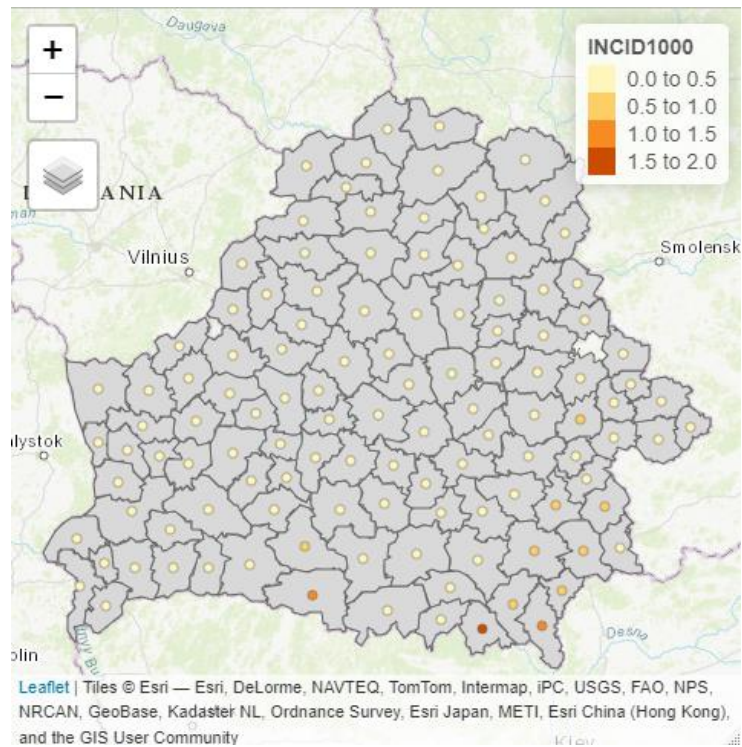
Limitations:

Thyroid cancer is seen in those individuals who were exposed to radiation when they were below 18 years at the time of Chernobyl disaster. Therefore, there is a potential for error in two ways. Firstly, the individuals rising in the high-risk zone in 1992(the year of data collection) may not be the individuals who were exposed in 1986. USSR initiated a mass mobilization post the Chernobyl disaster for a radius of nearly 60 km, which were later occupied once again leaving only a 30 km radius around the Chernobyl disaster site. Secondly, the soil Cs137 levels measured in the year 1992 may not accurately correlate with the amount of soil Cs137 or in fact the total exposure (since the individuals were also exposed to radiation through deposits for the air), as a significant amount of the radioactive material might be leached into nearby rivers or rain which might vary from one region to the other. Another significant limitation to this study is the authenticity of few datasets such as the ESRI datasets and there is no metadata and no CRS for the soil cesium 137 for the entire country of Belarus has not been mentioned, thus making it impossible to correlate the total pediatric thyroid cases with the soil cesium 137 levels for the country in a given year, 1992 in this case.

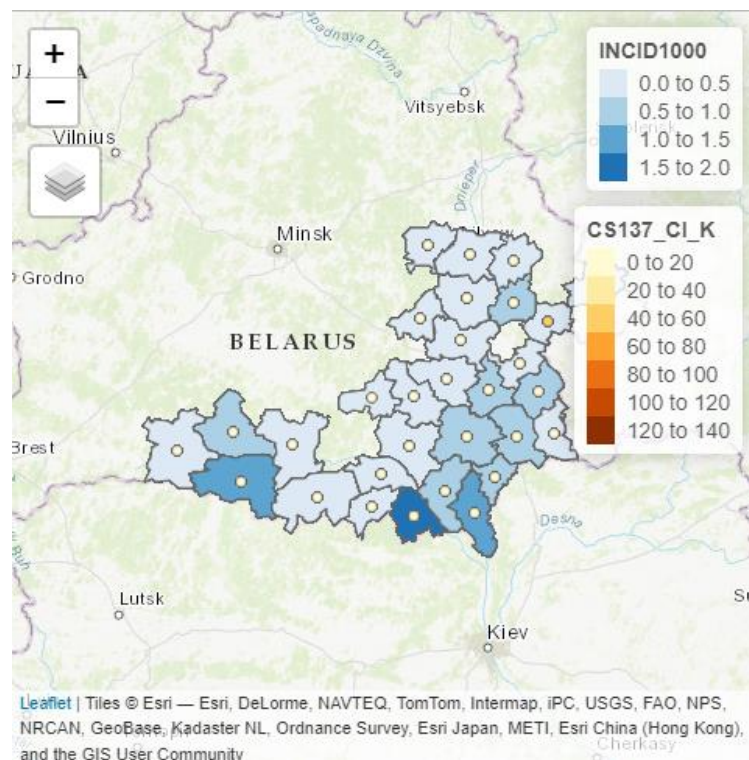
Maps:



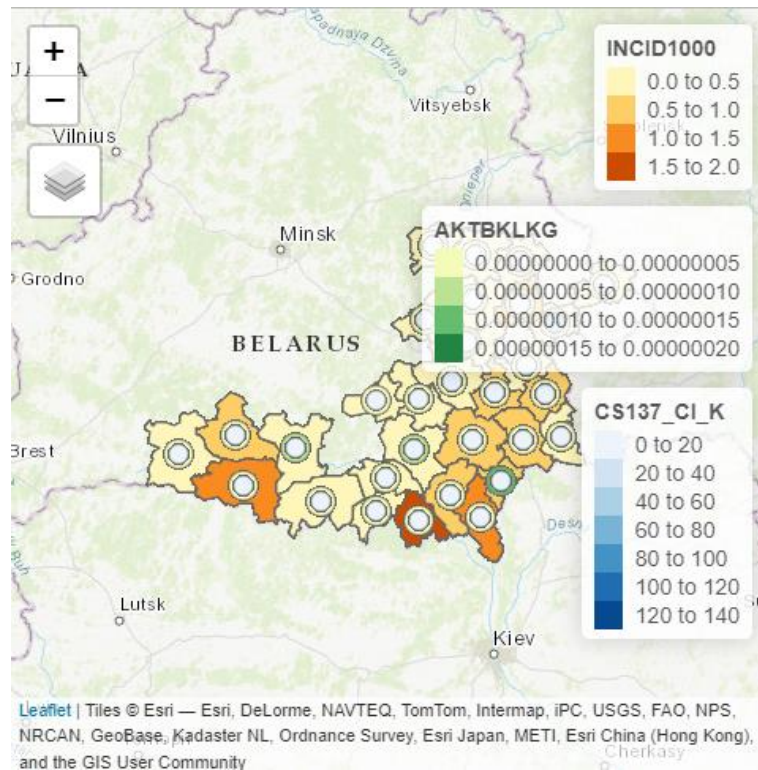
Map1: Soil Cs137 levels at 60km radius form the Chernobyl site(ci/km²)



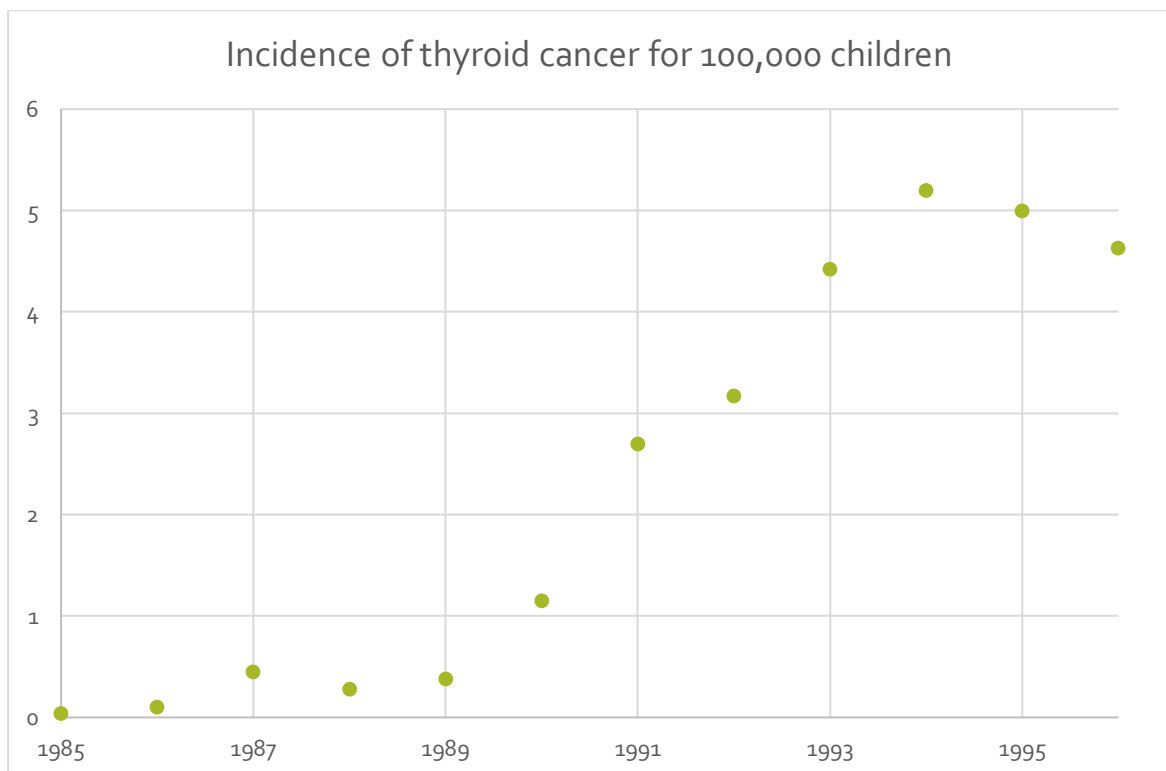
Map2: Incidence rates per 1000 children in the districts of Belarus(1992).



Map3: Incidence of pediatric thyroid cancer and soil Cs137 levels in the south western region of Belarus in 1992.(Incidence per 1000 children)



Map4: Cs137 levels in berries(curie per kg)with polygons representing the thyroid cancer incidence(per1000) and soil Cs137.



Plot1: Incidence rates of Pediatric thyroid cancer over the years.

Year	Girls	Boys	Total
1985			0.04
1986	0.14	0.07	0.1
1987	0.71	0.21	0.45
1988	0.28	0.27	0.28
1989	0.43	0.34	0.38
1990	1.13	1.17	1.15
1991	3.90	1.58	2.7
1992	3.54	2.82	3.17
1993	6.66	2.27	4.42
1994	7.30	3.16	5.2
1995	6.73	3.37	5.0
1996	6.38	2.95	4.63

Table 1: Average Incidence rates(per 100,000) of Pediatric Thyroid cancer in Belarus(1985 to 1996)

term	estimate	std.error	statistic	p.value	CI2.50%	CI97.50%
(Intercept)	-0.00896	0.196055	-0.0457	0.963642	-0.39802	0.380105
cs137	1.991017	0.034182	58.24732	7.56E-78	1.923184	2.05885

Table2: Linear model regression incidence rates of thyroid cancer and soil Cs137 levels.

term	estimate	std.error	statistic	p.value	CI2.50%	CI97.50%
(Intercept)	9.72E-09	1.33E-10	73.01683585	0	9.46E-09	9.99E-09
CS137_CI_K	1.49E-10	1.90E-11	7.830390509	5.17E-15	1.12E-10	1.86E-10

Table3: Linear regression model of the logCs137 levels in berries and logCs137 levels in the soil.

term	estimate	std.error	statistic	p.value	2.50%	97.50%
(Intercept)	0.616653296	0.057876855	10.65457503	7.43E-19	0.502010385	0.731296208
DISTANCE	-1.34E-06	1.74E-07	7.694972776	5.32E-12	-1.68E-06	-9.92E-07

Table4: Linear regression model of the soil Cs137 levels and the distance in km from the Chernobyl site.

References:

1. Chernobyl Appendix 1: Sequence of Events - World Nuclear Association [Internet]. [cited 2023 May 16]. Available from: <https://www.world-nuclear.org/information-library/safety-and-security/safety-of-plants/appendices/chernobyl-accident-appendix-1-sequence-of-events.aspx>
2. chernobyl.pdf [Internet]. [cited 2023 May 16]. Available from: <https://www.iaea.org/sites/default/files/chernobyl.pdf>
3. Chernobyl | Chernobyl Accident | Chernobyl Disaster - World Nuclear Association [Internet]. [cited 2023 May 16]. Available from: <https://world-nuclear.org/information-library/safety-and-security/safety-of-plants/chernobyl-accident.aspx>
4. Fact Sheet 0902 - Cesium-137 | Ohio Department of Health [Internet]. [cited 2023 May 16]. Available from: <https://odh.ohio.gov/know-our-programs/radiological-health-and-safety/media/behrrp-fact-sheet-0902>
5. <https://plus.google.com/+UNESCO>. Belarus: Facing the disaster alone [Internet]. UNESCO. 2021 [cited 2023 May 16]. Available from: <https://en.unesco.org/courier/october-2000/belarus-facing-disaster-alone>
6. Gray R. The true toll of the Chernobyl disaster [Internet]. [cited 2023 May 16]. Available from: <https://www.bbc.com/future/article/20190725-will-we-ever-know-chernobyls-true-death-toll>
7. Chernobyl May Have Been Gorbachev's Greatest Lesson [Internet]. The Cipher Brief. [cited 2023 May 16]. Available from: https://www.thecipherbrief.com/column_article/chernobyl-may-have-been-gorbachevs-greatest-lesson

Appendix:

1. Incidence rates of pediatric thyroid cancer:
https://inis.iaea.org/collection/NCLCollectionStore/_Public/29/017/29017355.pdf
https://inis.iaea.org/collection/NCLCollectionStore/_Public/31/056/31056912.pdf
2. Soil Cs137, Berry Cs137, Chernobyl site locations:

<https://www.esri.com/news/arcnews/fall03articles/analyzing.html>

<https://desktop.arcgis.com/fr/analytics/case-studies/an-appropriate-kriging-model-for-soil-contamination.htm>

3. Soil Cs137 levels at 60km radius from the Chernobyl site:

<https://eidc.ac.uk/>

R code:

```
library("rlang")
library("GSODR")
library(foreach)
library(tidyverse)
library("stars")
library(tmaptools)
library(tmap)
library(tidycensus)
library("tidymodels")
library("readxl")
library("vip")
library(sf)
library(rgdal)
library(spaMM)
library(stars)
library(spaMM)
library(leaflet)
library(leaflet)
library(leaflet)
library(elevatr)
library(lubridate)
library(dplyr)
library(raster)
```

```

#Getting the chernobyldata into R environment ##

setwd("G:\\CSPH\\R Geocomputation\\final project")

sample7 <- st_read("assignment A1.1")
st_layers("assignment A1.1")
##Berries data##
sample7berry <- st_read("assignment A1.1", layer = "BerryCs")

conversion_factor <- 3.7e10

# Convert BerryCs from Bq/kg to Ci/kg
sample7berry$AKTBKLG <- sample7berry$AKTBKLG / conversion_factor

##Belarus border data-southwest region##
sample7border <- st_read("assignment A1.1", layer = "Border_Cs137")
##Chernobyl site pointdata##
sample7chernobyl <- st_read("assignment A1.1", layer = "Chernobyl")
##Settlement areas data##
sample7settlements <- st_read("assignment A1.1", layer = "settlements")

##Cs137data##
sample7Cs137 <- st_read("assignment A1.1", layer = "Cs137")

##thyroid cancer incidence datasouthwest##
sample9 <- st_read("Thyroidcancer")
sample9_valid <- st_make_valid(sample9)

##Thyroid cancer whole Belarus##
sample10 <- st_read("thyroid2")
sample10_valid <- st_make_valid(sample10)

##Cs137 in Ukraine data##
chernobyl_data1 <- read.csv("G:\\CSPH\\R Geocomputation\\final
project\\chernobyldata\\1_Spatial_dataset.csv")

chernobyl_sf <- st_as_sf(chernobyl_data1, coords = c("Longitude", "Latitude"), crs = 4326)

chernobyl_transformed <- st_transform(chernobyl_sf, st_crs(sample10_valid))

##60km radius data##
chernobyl_data2 <- read.csv("G:\\CSPH\\R Geocomputation\\final
project\\chernobyldata\\dep_60km_1987.csv")

chernobyl_60 <- chernobyl_data2 %>% mutate(distance = Distance_from_ChNPP_km *
1000)

## Converting the angle and distance from the Chernobyl site into coordinates##

```

```

chernobyl_coords <- st_coordinates(sample7chernobyl)
chernobyl_x <- chernobyl_coords[1]
chernobyl_y <- chernobyl_coords[2]

calculate_coordinates <- function(chernobyl_x, chernobyl_y, angle, distance) {
# Converting angle from degrees to radians
angle_radians <- angle * (pi / 180)

# Calculating the coordinates
x <- chernobyl_x + distance * sin(angle_radians)
y <- chernobyl_y + distance * cos(angle_radians)

return(c(x, y))
}

chernobyl_60$X <- NA
chernobyl_60$Y <- NA

for (i in 1:nrow(chernobyl_60)) {
angle <- chernobyl_60[i, 2] # Angle is in the second column
distance <- chernobyl_60[i, 22] # distance is in the 22nd column

# Calculating the coordinates using the formula
coordinates <- calculate_coordinates(chernobyl_x, chernobyl_y, angle, distance)

# Storing the calculated coordinates in the dataset
chernobyl_60[i, "X"] <- coordinates[1]
chernobyl_60[i, "Y"] <- coordinates[2]
}
##Converting into sf object with crs##
chernobyl_60t <- st_as_sf(chernobyl_60, coords = c("X", "Y"), crs =
st_crs(chernobyl_transformed))

# Conversion factor from Bq/m^2 to Ci/km^2
curie_per_km2_conversion_factor <- 2.7027e-5

# Creating a new column for Cs-137 levels in curie per kilometer square
chernobyl_60t$Cs137_Ci <- chernobyl_60t$Cs.137_Bqm2 *
curie_per_km2_conversion_factor

# Plotting the data with the new column
tm_basemap("Esri.WorldTopoMap") +

```

```

tm_shape(chernobyl_60t) +
tm_dots("Cs137_Ci")+
tm_shape(sample7chernobyl)+
tm_dots()

```

##sample10 dataset contains the incidence of thyroid cancer for the year 1992 all over belarus##

##It also has details on the distance of each point from the chernobyl reactor. The steps##

##here outline the mapping of the values and running a linear model to see if any correlation##

##exists between the distance from the chernobyl reactor and the incidence rate of thyroid cancer##

```

fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(sample10_valid) +
  tm_polygons()
tmap_mode("view")
tmap_options(check.and.fix = TRUE)
tm_basemap("Esri.WorldTopoMap") +
tm_shape(sample10_valid) +
tm_polygons() +
  tm_shape(sample10_valid) +
  tm_dots("INCID1000")

```

```

model8 <- lm(INCID1000 ~ DISTANCE, data = sample10_valid)
summary(model8)

```

```

fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(chernobyl_transformed) +
  tm_dots("X137Cs")

```

```

library(raster)
library(tmap)

```

```

chernobyl_dataraster <- raster(chernobyl_transformed)

```

```
chernobyl_dataraster <- rasterize(chernobyl_transformed, chernobyl_dataraster, field =  
"X137Cs")
```

```
fproj1 <- tmap_mode("view")  
tm_basemap("Esri.WorldTopoMap") +  
  tm_shape(chernobyl_dataraster)+  
  tm_raster()
```

```
sample9_valid <- st_make_valid(sample9)  
# Plot the polygons(just basic district polygons for the whole Belarus) ##
```

```
sample9_valid <- st_buffer(sample9, 0)
```

```
tm_basemap("Esri.WorldTopoMap") +  
  tm_shape(sample9_valid) +  
  tm_polygons()
```

```
##Calculating the expected counts using probmap from sdep package##  
library(spdep)  
pmap <- probmap(sample9_valid$CASES, sample9_valid$POPULUSE)
```

```
pmapjoined <- cbind(sample9_valid, pmap)
```

```
fproj1 <- tmap_mode("view")  
tm_basemap("Esri.WorldTopoMap") +  
  tm_shape(pmapjoined) +  
  tm_dots("relRisk")
```

```
fproj1 <- tmap_mode("view")  
tm_basemap("Esri.WorldTopoMap") +  
  tm_shape(pmapjoined) +  
  tm_dots("expCount")
```

```
##Plotting the berry data##
```

```
library(tmap)  
library(tmaptools)  
fproj1 <- tmap_mode("view")  
tm_basemap("Esri.WorldTopoMap") +  
  tm_shape(sample7berry)+  
  tm_dots("AKTBKLGK")
```

```
fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(sample7border)+
  tm_polygons()
```

```
fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(sample7chernobyl)+
  tm_dots()
```

```
fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(sample7Cs137)+
  tm_dots("CS137_CI_K")
```

```
fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(sample7settlements)+
  tm_dots()
```

```
fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(sample9_valid)+
  tm_polygons("INCID1000")
```

##joining the thyroid cases data with the southwest region cs137 at the intersections##

```
intersect_layer <- st_intersection(pmapjoined, sample7Cs137)
```

```
fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(pmapjoined)+
  tm_polygons("INCID1000")+
  tm_shape(intersect_layer)+
  tm_dots("CS137_CI_K")
```

```
model1 <- lm(INCID1000 ~ CS137_CI_K, data = intersect_layer)
```

```
model1 <- lm(INCID1000 ~ CS137_CI_K, data = intersect_layer)
summary(model1)
```

```
model2 <- lm(relRisk ~ CS137_CI_K, data = intersect_layer)
summary(model2)
```

```
##joining the berries data with thyroid cancer polygons##
intersect_layer2 <- st_join(sample7berry, pmapjoined)
```

```
st_crs(sample7berry)
st_crs(sample7Cs137)
```

```
library(gstat)
library(sp)
library(sf)
```

```
model4 <- lm(INCID1000 ~ AKTBKLKG, data = intersect_layer2)
summary(model4)
```

```
##joining pmap(thyroid data) with cs137 data and keeping the polygons. Then joining the ##
##berry data with pmap data keeping the polygons and then joining both of them with ##
##geometry. ##
```

```
join1 <- st_join(pmapjoined, sample7Cs137, join = st_contains, keep = TRUE)
```

```
fproj1 <- tmap_mode("view")
tm_basemap("Esri.WorldTopoMap") +
  tm_shape(join1)+
  tm_polygons("INCID1000")+
  tm_shape(join1)+
  tm_dots("CS137_CI_K")
```

```
join2 <- st_join(join1, sample7berry, join = st_contains, keep = TRUE)
```

```
join3 <- na.omit(join2)
join3 <- st_make_valid(join3)
```

```
fproj1 <- tmap_mode("view")

tm_basemap("Esri.WorldTopoMap") +
  tm_shape(join3) +
  tm_polygons("INCID1000") +
  tm_shape(join3)+
```

```
tm_dots("AKTBKLG", palette = "YlGn", size = 0.2 )+  
tm_shape(join3)+  
tm_dots("CS137_CI_K", palette = "Blues", size = 0.1)
```

```
model7 <- lm(log_berry ~ log_cs137, data = join3)  
summary(model7)
```

```
model1 <- lm(INCID1000 ~ CS137_CI_K, data = intersect_layer)
```

```
model1 <- lm(INCID1000 ~ CS137_CI_K, data = intersect_layer)  
summary(model1)  
model1conf <- confint(model1)  
write.csv(model1conf, file = "model1_conf.csv", row.names = FALSE)
```

```
model2 <- lm(relRisk ~ CS137_CI_K, data = intersect_layer)  
summary(model2)
```

```
model4 <- lm(INCID1000 ~ AKTBKLG, data = intersect_layer2)  
summary(model4)
```

```
model7 <- lm(AKTBKLG ~ CS137_CI_K, data = join3)  
summary(model7)  
model7conf <- confint(model7)  
write.csv(model7conf, file = "model7_conf.csv", row.names = FALSE)
```

```
model7_summary <- tidy(model7)  
write.csv(model7_summary, file = "model7_summary.csv", row.names = FALSE)
```

```
library(broom)  
model1_summary <- tidy(model1)  
write.csv(model1_summary, file = "model1_summary.csv", row.names = FALSE)
```

```
model2_summary <- tidy(model2)  
write.csv(model2_summary, file = "model2_summary.csv", row.names = FALSE)
```

```
model8_summary <- tidy(model8)  
write.csv(model8_summary, file = "model8_summary.csv", row.names = FALSE)  
model8conf <- confint(model8)  
write.csv(model8conf, file = "model8_conf.csv", row.names = FALSE)
```

```
model4_summary <- tidy(model4)  
write.csv(model4_summary, file = "model4_summary.csv", row.names = FALSE)
```